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Paper

BiGGViz: A MATLAB App for Visualization of Genome-Scale BiGG-Based Metabolic Networks

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Abstract

Motivation: Genome-scale metabolic reconstructions provide curated, constraint-based models of cellular metabolism, linking genes, reactions, and metabolites in a unified stoichiometric framework. The Biochemical, Genetic and Genomic knowledge base (BiGG) distributes such reconstructions with standardized identifiers, facilitating model reuse and integration with computational ecosystems such as the COBRA Toolbox. Accessible visualization of the resulting reaction and metabolite networks is essential to make these large models interpretable, communicate analyses, and support users in exploring metabolic structures
Results: BiGGViz is a MATLAB toolbox for metabolic network visualization that takes COBRA-compatible BiGG models as input and automatically constructs reaction-reaction and metabolite-metabolite graphs for interactive exploration. The app provides a graphical interface for loading models, switching between reaction and metabolite views, tuning node appearance, applying currency-metabolite and topology-based filters, inspecting local neighborhoods, highlighting pathways with configurable hop-based expansion, and exporting the current view or network structure for downstream usage. Optional annotation tables allow users to overlay user-defined node appearance and COBRA-derived analyses—such as flux balance analysis (FBA) fluxes, enabling flux-aware inspection of dense genome-scale networks. BiGGViz supports exploratory analyses, result communication, and classroom demonstrations for both coding and non-coding users, while remaining compatible with Cytoscape-based downstream workflows. A scriptable wrapper, `exportBiGGViz`, is also available for code-driven pipelines.
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Keywords metabolic models, BiGG, COBRA, network analysis, visualization

Abbreviations BiGG, COBRA, are these abbr of above keywords?

Graphical Abstract

Do we need a graphical abstract?

Key Messages

- Do we need key messages?

Introduction

Logical flow: reconstruction exist → COBRA turns them into computable models → BiGG provides curated models ready for COBRA → why visualization is needed downstream.

Genome-scale metabolic reconstructions compile the known metabolic reactions, metabolites, and their biochemical relationships for an organism into a coherent stoichiometric framework, providing a structured representation of cellular metabolism that supports quantitative, constraint-based modeling. Within the COnstraint-Based Reconstruction and Analysis (COBRA) framework, these reconstructions are formulated as stoichiometric models subject to physicochemical

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and environmental constraints, enabling a broad family of analyses, including prediction of steady-state flux distributions, assessment of pathway usage under different conditions, simulation of gene or reaction knockouts, and integration of omics data to build context-specific models. Together, these methods allow researchers to interrogate metabolic capabilities in silico across organisms, media and perturbations.

The Biochemical, Genetic and Genomic knowledge base (BiGG) provides a curated collection of such genome-scale reconstructions with standardized reaction and metabolite identifiers, consistent annotations, and distribution formats that are directly compatible with COBRA toolboxes. In typical workflows, BiGG models are imported to the COBRA Toolbox or related software, where users perform flux balance analysis (FBA), flux variability analysis (FVA), reaction or gene deletion studies, and context-specific model derivation, among other tasks. Regardless of the specific analysis, interpreting these outputs generally requires understanding how predicted fluxes, blocked reactions, or pruned subnetworks map onto the underlying reaction or metabolite network structure. This is particularly challenging at the genome scale, where models may comprise hundreds to thousands of reactions and metabolites.

A variety of tools support metabolic modeling and network visualization, but each addresses a different need and leaves gaps for users working directly with BiGG/COBRA models in MATLAB. BiGGR focuses on constraint-based modeling and flux visualization in R, emphasizing flux distributions on relatively small metabolic models and pathway diagrams rather on general-purpose reaction and metabolic graphs for arbitrary COBRA reconstructions. MATLAB-based toolboxes such as SBEToolbox and the Metabolite Network Toolbox concentrate on network topology or kinetic and thermodynamic modeling, often operating on user-defined adjacency matrices or SBML models instead of treating standardized BiGG/COBRA model structs as their primary input. Other systems, including MetaboNetworks and web components like MetExportViz, provide rich exploration of curated pathways or metabolite graphs tied to specific resources (e.g. KEGG) or dedicated web back ends, but they are not designed as MATLAB-native tools that sit on top of BiGG/COBRA and directly consume their model structs. General-purpose platforms such as Cytoscape offer powerful, extensible environments for graph analysis and visualization, yet they require exporting networks and data from COBRA into exchange formats and working outside the MATLAB environment. Methods like GIMME and related algorithms address a different stage of the pipeline by generating context-specific models from expression data, while leaving visualization to downstream tools.

From the perspective of a BiGG/COBRA user working in MATLAB, there is therefore a practical gap: there is no lightweight, COBRA-aware visualization layer that directly accepts BiGG model structs and, without additional graph-construction code, produce interpretable reaction-reaction and metabolite-metabolite network views that can be explored interactively and annotated with analysis results. BiGGViz is designed to fill this niche by treating BiGG-derived COBRA models as its primary input, interpreting the stoichiometric matrix as a bipartite reaction-metabolite graph, and rendering separate, interactive graphs for exploration and annotation. The core contribution is a MATLAB App that enables users to load BiGG/COBRA models, choose between reaction- and metabolite-centric views, adjust visual encodings, apply currency-metabolite and topology-based filters, explore local neighborhoods and pathways with hop-based expansion, and export the current view or network structure for downstream analysis and communication. Optional annotation tables allow users to overlay attributes such as pathway labels, compartments, gene associations, or COBRA-derived quantities—including, when available, FBA fluxes—so that these analyses can be inspected within the context of the underlying network without requiring additional scripting. For users who prefer code-driven pipelines, a scriptable wrapper, `exportBiGGViz`, exposes the same graph constructing core and supports use cases that complement the app, such as batch generation of network viewers across multiple models, automated integration into reproducible analysis pipelines, and programmatic control over filtering parameters and annotation overlays.

Implementation

Software Overview

BiGGViz is implemented as a lightweight visualization layer on top of COBRA-style metabolic models in MATLAB. The core of the toolbox consists of functions that take a COBRA-compatible model struct as input and derive corresponding reaction-reaction and metabolite-metabolite networks from the underlying stoichiometric matrix S. This graph constructing layer is primarily driven through the BiGGViz MATLAB App, which provides an interactive graphical environment for exploratory analysis.

A key design choice is to keep the network structure and annotation metadata logically separate. The BiGG/COBRA model defines the stoichiometric network and provides core annotations such as names, subsystems, and formulae. Optional annotation tables allow users to augment or override these fields with additional attributes, such as custom labels, FBA fluxes, node or edge sizes. These are merged ino the visualization as node attributes without altering the underlying model. Figure 1 summarizes the overall pipeline from a BiGG/COBRA model and annotation table to derived networks, interactive viewers, and exported artifacts.

Data model and network construction

BiGGViz interprets the stoichiometric matrix S, together with reaction and metabolite identifiers (`rxns`, `mets`), as a basis for constructing separate reaction-centric and metabolite-centric networks. For the reaction network, each reaction in `model.rxns` is represented as a node, and an undirected edge is added between two reactions when they share at least one metabolite (i.e., the reaction-layer projection of the underlying bipartite reaction-metabolite graph). For the metabolite network, each metabolite in `model.mets` is a node, and two metabolites are connected when they co-participate in at least one reaction. Edges are treated as undirected in both views; reversible and irreversible reactions are distinguished via annotations (e.g., reversibility flags or flux values) rather than

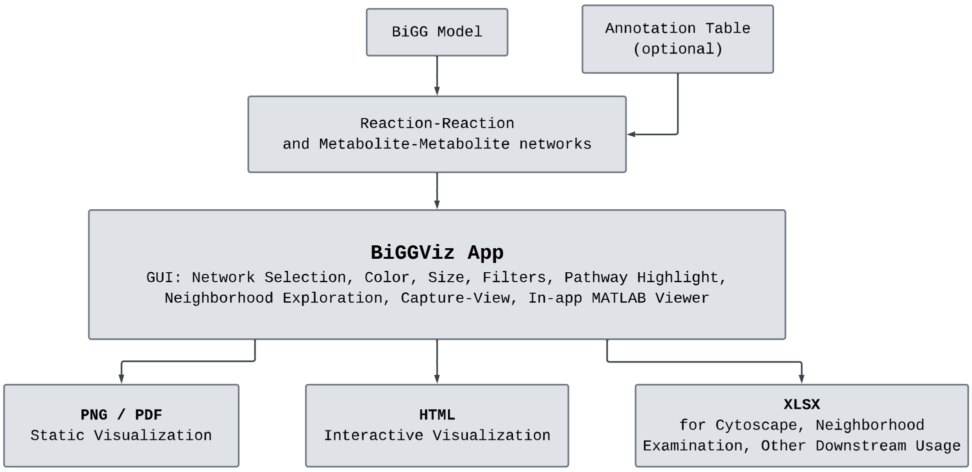

Figure 1 Overview of the BiGGViz pipeline.

directed edges, reflecting that the primary goal is structural exploration rather than simulation.

Genome-scale and other large reconstructions often contain highly connected "currency" metabolites, such as water, ATP, or ubiquitous cofactors, which form dense hubs and can obscure pathway structure. To mitigate this, BiGGViz includes an optional currency-metabolite filtering mechanism controlled by a boolean flag (`filtercurrency`) and a degree threshold (`currencyDegThresh`), which removes nodes exceeding a user-specified connectivity cutoff prior to visualization. This heuristic reduces clutter and highlights pathway-level structure at the cost of omitting generic links; users can disable the filter to recover the full network when needed. To remain scalable to large constructions, BiGGViz relies on sparse matrix operations to derive adjacency relationships from S, and constructs only the requested network type(s), further complemented by in-app controls such as maximum node degree and minimum component size filters to keep dense views responsive and interpretable.

BiGGViz App: interactive MATLAB interface

The BiGGViz App is implemented in MATLAB App Designer and wraps the graph constructing layer in an interactive graphical interface. Users can load a BiGG model from `.mat` or SBML/XML files and, optionally, an annotation table in `.csv` or `.xlsx` format, which is standardized via a `parseAnnotations` routine to ensure that reaction or metabolite identifiers align with the underlying model. The interface comprises a control panel for data loading and display options, a central visualization panel for the current network view, and auxiliary panels for legends and export actions.

Within the app, users can switch between reaction-centric and metabolite-centric modes, choose among several color schemes (uniform, pathway/subsystem, reversibility, gene association, FBA flux, compartment, or metabolite class), and select node size mappings (uniform, degree-based, or flux-based). Structural filters, including maximum node degree and minimum component size, simplify dense networks in combination with the currency-metabolite filter to suppress dense hubs or small fragments and make pathway-level structure more apparent. Neighborhood exploration tools support both node-centered and pathway-seeded exploration with 1-3 hop expansion, allowing users to trace how a selected node or pathway connects to its surroundings. A pathway-highlight dropdown enables users to retain a pathway of interest while dimming adjacent nodes, and a search box provides direct navigation to nodes by identifier. A "Capture View" control records the current layout as a static graphic.

At any point, the current view can be exported directly from the app as an interactive HTML viewer, as static PNG or PDF figures tuned for print-friendly layouts, or as a Cytoscape-oriented Excel workbook containing separate node and edge tables for downstream analysis. These export options make it straightforward to transition from exploratory visualization inside MATLAB to presentation-ready figures and further network analysis in external platforms.

Implementation and validation

BiGGViz is implemented in MATLAB and depends on the COBRA Toolbox for model representation, input/output routines, and optional flux analysis. It operates on COBRA-compatible model structs, including BiGG models distributed in `.mat` or SBML formats and user-specific reconstructions loaded via functions such as `readCbModel` or `loadBiggModel`. Development and testing have been carried out on Windows, but the code itself is platform-independent aside from MATLAB and COBRA availability; users enable the toolbox by adding the BiGGViz folder to the MATLAB path and initializing the COBRA Toolbox in their session.

For users who prefer code-driven workflows, BiGGViz also provides a lightweight scriptable wrapper, `exportBiGGViz`, which invokes the same graph-construction routines used by the app. Given a COBRA-compatible model and optional annotation table, `exportBiGGViz` generates interactive HTML viewers for reaction and/or metabolite networks directly from a MATLAB script, enabling batch processing of multiple models and integration of visualization steps into reproducible COBRA analysis pipelines without manual interaction in the GUI.

BiGGViz has been validated on exemplar BiGG models spanning distinct organisms and reconstruction sizes, including the E. coli core model, a human red blood cell model, and a Methanosarcina barkeri model, and further tested on a panel of 50 curated BiGG reconstructions in both .mat and SBML formats. Across these cases, the app and scriptable wrapper consistently loaded models, constructed the requested network views, and produced the expected interactive outputs, with particular attention paid to annotation handling, the behavior of filtering controls, and flux-aware visualization when FBA results were available.

Results

Example datasets and loading workflows

To illustrate BiGGViz on representative BiGG/COBRA workflows, we applied the app to the *E. coli* core model (`e_coli_core`). The model was downloaded directly from BiGG Models database as a `.mat` or SBML/XML file, loaded via the COBRA Toolbox, after which BiGGViz constructed reaction- and metabolic-centric networks without additional graph constructing code.

Network views and controls available for the E. coli core model

Figure 2 illustrates BiGGViz applied to the *E. coli* core model and contrasts the resulting network views with a reference Escher pathway map downloaded from the Escher website. This panel illustrates different aspects of BiGGViz functionality on the reaction- and metabolite-centered graphs, multiple node encodings, and comparison to an established pathway map—demonstrating that whole network views can be generated and re-encoded without manual redrawing.

The app's neighborhood and pathway tools enable users to move from global views to focused local analyses. For the *E. coli* core model, we illustrate a one-hop metabolite neighborhood around glucose-6-phosphate (g6p) in the metabolite network, with nodes colored by metabolite class and labels displayed for all nodes in the subgraph; this view makes it straightforward to inspect immediate biochemical neighbors of a focal metabolite and their functional categories. In the reaction network, applying maximum-degree and minimum-component-size filters yields a sparser graph in which reactions are colored by gene-association status, separating reactions with known gene associations from those lacking them and making it easier to identify poorly annotated regions of the network. Finally, an one-hop reaction neighborhood around the Citric Acid Cycle pathway, with nodes colored and labeled by pathway, demonstrates how the pathway highlight and hop-expansion controls can be used to examine cross-talk between a canonical pathway and its immediate neighbors in the network. Collectively, these examples show how BiGGViz supports both global exploration and targeted inspection of specific metabolites, reactions, or pathways within the same interface.

To demonstrate the use of BiGGViz beyond curated BiGG models, we include an example user-supplied reconstruction together with a companion annotation table. The model file provides the stoichiometric structure in the standard BiGG/COBRA layout, while the annotation table (Table 1) lists 10 reactions with aligned reaction identifiers and columns such as names, pathway labels (e.g. glycolysis, fermentation, pentose phosphate pathway), reaction formulae, enzyme names, user-defined node sizes and FBA flux values. When this table is loaded via the app alongside the model, BiGGViz automatically merges the annotations onto the corresponding nodes, enabling users to color reactions by pathway or category, scale node size by the NodeSize column, and optionally overlay flux information from the FBAFlux column without modifying the underlying stoichiometric network. This example illustrates how BiGGViz can serve as a general visualization front end for both BiGG-curated and user-specific COBRA models, and how simple spreadsheet-style annotation tables can expose custom attributes and analysis outputs in the interactive network views.

Conclusion

Some Conclusions here.

Conflicts of interest

The authors declare that they have no competing interests.

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Data availability

The data underlying this article are available in [repository name, eg, the GenBank Nucleotide Database] at [URL], and can be accessed with [unique identifier, eg, accession number, deposition number].

Author contributions statement

Must include all authors, identified by initials, for example: S.R. and D.A. conceived the experiment(s), S.R. conducted the experiment(s), S.R. and D.A. analysed the results. S.R. and D.A. wrote and reviewed the manuscript.

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